ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2023

B.TECH. CSE ENGINEERING — SEMESTER 4

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Describe OS structures and Process Control Block (PCB) state transitions. [14 Marks]

[OR]

Q.1 (Alt) Explain system calls fork(), exec(), wait(), and exit(). Compare User vs Kernel threads. [14 Marks]

--- UNIT II ---

Q.2 Calculate turnaround and waiting times for Round Robin and SJF. Define Critical Section. [14 Marks]

[OR]

Q.2 (Alt) Explain Peterson's solution. Solve Producer-Consumer problem using Semaphores. [14 Marks]

--- UNIT III ---

Q.3 State Coffman conditions for deadlock. Demonstrate Banker's Algorithm with Safety check. [14 Marks]

[OR]

Q.3 (Alt) Describe Deadlock Detection, Recovery, and Resource Allocation Graphs (RAG). [14 Marks]

--- UNIT IV ---

Q.4 Explain Paging and TLB effective access time. Calculate page faults for FIFO, LRU, Optimal. [14 Marks]

[OR]

Q.4 (Alt) Explain Belady's Anomaly and Segmentation address translation hardware. [14 Marks]

--- UNIT V ---

Q.5 Compare Disk Scheduling algorithms: FCFS, SSTF, SCAN, C-SCAN. Explain File Allocation. [14 Marks]

[OR]

Q.5 (Alt) Describe RAID levels (0, 1, 5, 6) and Thrashing Working Set model. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****